

# CSE460: VLSI Design

## Lecture 1

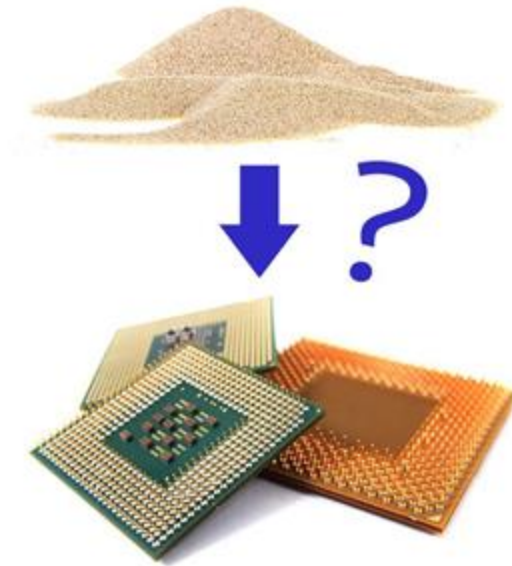
**Very Large Scale Integration or VLSI**

# Contents

- Topics of this course
- History
- Transistors
- Moore's law
- Types of chips
- Design abstraction
- Design methodology

# What will you learn in this course?

- You may have heard people say that electronics are made from sand, but do you know how?



# What will you learn in this course?

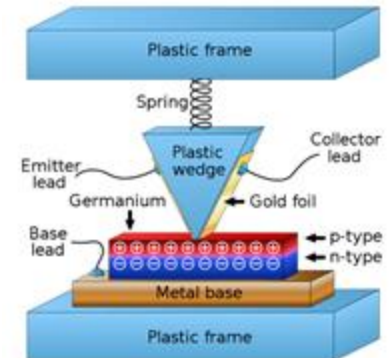
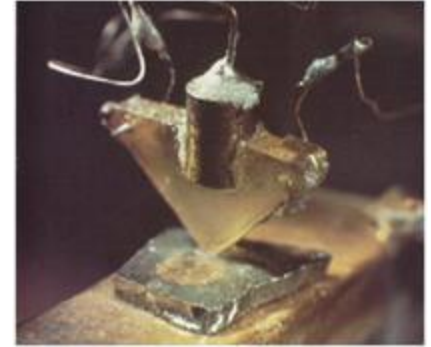
- A solid introduction to VLSI
- Different kinds of chips/ICs
- How to build one
- Theoretical background to making a good chip
- Quality measures of a chip
- The complete lifecycle of a chip

# The Transistor

- Transistor concept
  - Transistors can be viewed as **electrically controlled switches** with a control terminal and two other terminals that are **connected** or **disconnected** depending on the *voltage* or *current* applied to the control terminal

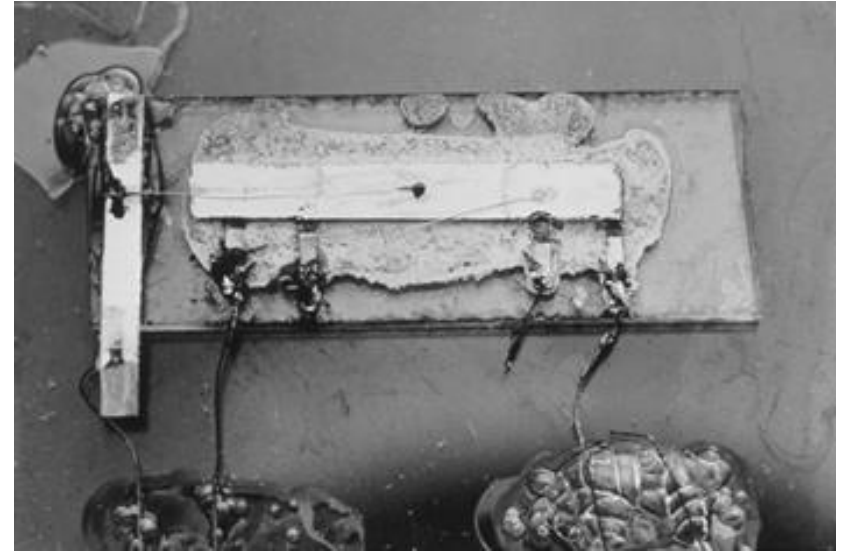
# Transistor Invention

- 1947: First transistor was built
  - John Bardeen and Walter Brattain built the first functioning point contact transistor at Bell Labs
  - Nearly classified as military secret
  - Later publicly introduced by Bell Labs
  - **Nobel Prize in Physics in 1956 for Bardeen, Brattain, and their supervisor William Shockley**
  - *We have called it the Transistor, T-R-A-N-S-I-S-T-O-R, because it is a resistor or semiconductor device which can amplify electrical signals as they are transferred through it from input to output terminals. It is, if you will, the electrical equivalent of a vacuum tube amplifier. But there the similarity ceases. It has no vacuum, no filament, no glass tube. It is composed entirely of cold, solid substances.*

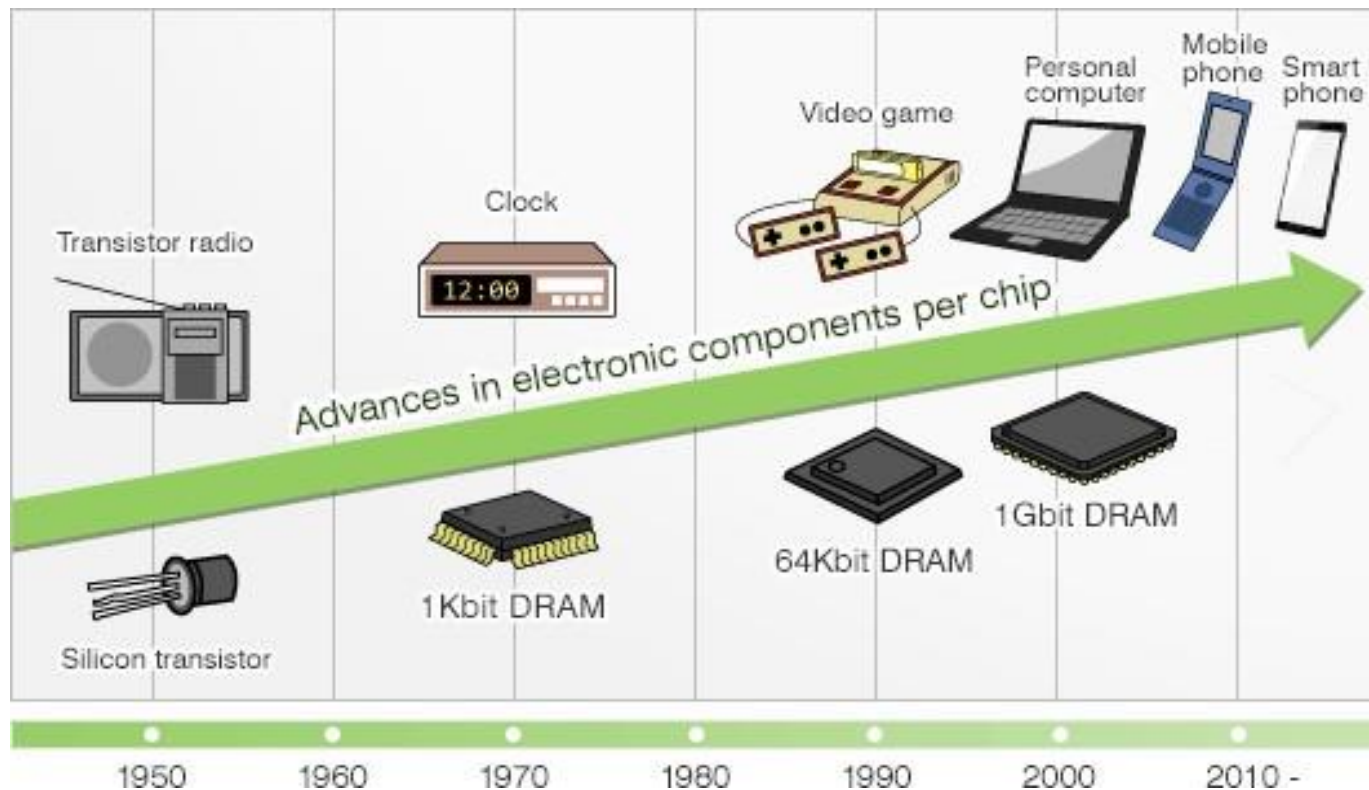


# First IC

- 1959: U.S. Patent 3,138,743 for "Miniaturized Electronic Circuits"
  - Several years later after the invention of transistors, Jack Kilby at Texas Instruments realized the potential for miniaturization if multiple transistors could be built on one piece of silicon
  - **Kilby received the Nobel Prize in Physics in 2000 for the invention of the integrated circuit**



# Evolution of Electronic Devices



# Evolution of Electronic Devices



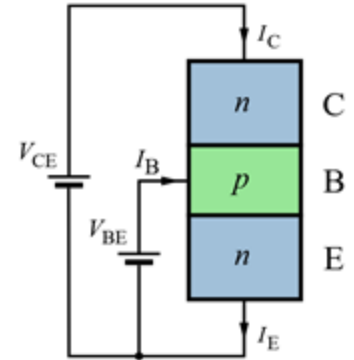
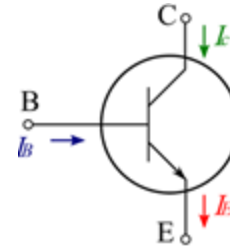
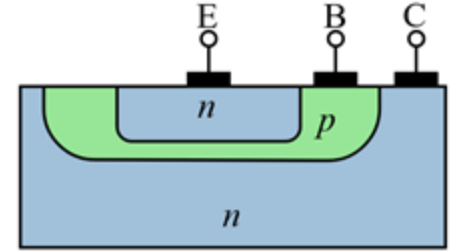
5MB (IBM)-1956, Cost - \$10,000/ 1 MB



4TB (WB)-2013

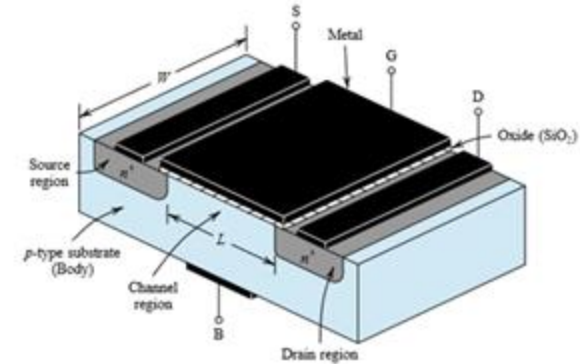
# Bipolar Junction Transistor

- Bipolar Junction Transistor (BJT)
  - Developed by Bell Labs
- More reliable, less noisy, and more power-efficient than the point contact transistor
- BJTs require a small current into the control (*base*) terminal to switch much larger currents between the other two (*emitter* and *collector*) terminals
- **Power being dissipated by these base currents even when idle**



# Metal Oxide Semiconductor Field Effect Transistor

- 1960s: (MOSFETs) began to enter production
- Metal Oxide Semiconductor Field Effect Transistors or MOSFETs offer the compelling advantage that they draw almost zero control current while idle.
- Two flavors: nMOS and pMOS
- Three terminals: *Gate* (control), *Drain* and *Source*



# First MOSFET Gates

- 1963: First logic gate using MOSFETs
  - Frank Wanlass at Fairchild described the first logic gates using MOSFETs
- Fairchild's gates used both nMOS and pMOS transistors, earning the name Complementary Metal Oxide Semiconductor, or CMOS
- The circuits used discrete transistors but consumed six orders of magnitude less power than their bipolar counterparts
- With the development of the silicon planar process, MOS integrated circuits became attractive for their low cost because each transistor occupied less area and the fabrication process was simpler

## Integration Levels

**SSI:** 1-12 gates

**MSI:** 13-99 gates

**LSI:** 100-9999 gates

**VLSI:** 10-99.9 k gates

**ULSI:** >100k gates

# Integration Levels

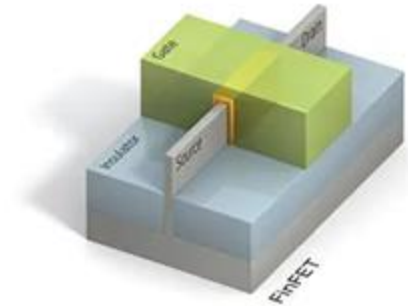
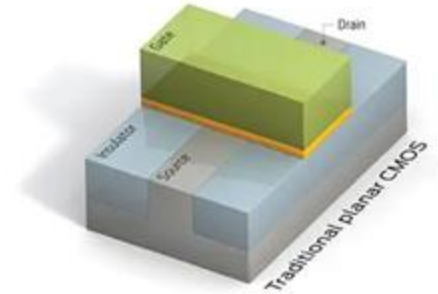
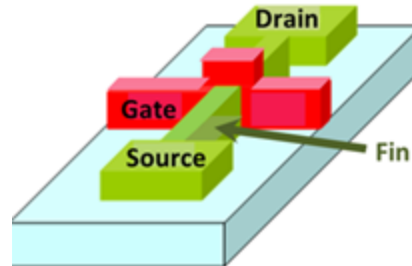
| Name                                 | No of Transistors   | No of Logic Gates |
|--------------------------------------|---------------------|-------------------|
| Small-Scale Integration (SSI)        | 1 to 10             | 1 to 12           |
| Medium-Scale Integration (MSI)       | 10 to 500           | 13 to 99          |
| Large-Scale Integration (LSI)        | 500 to 20,000       | 100 to 9,999      |
| Very Large Scale Integration (VLSI)  | 20,000 to 1,000,000 | 10,000 to 99,999  |
| Ultra Large Scale Integration (ULSI) | 1,000,000 and more  | 100,000 and more  |

# Trend in MOSFET IC Design

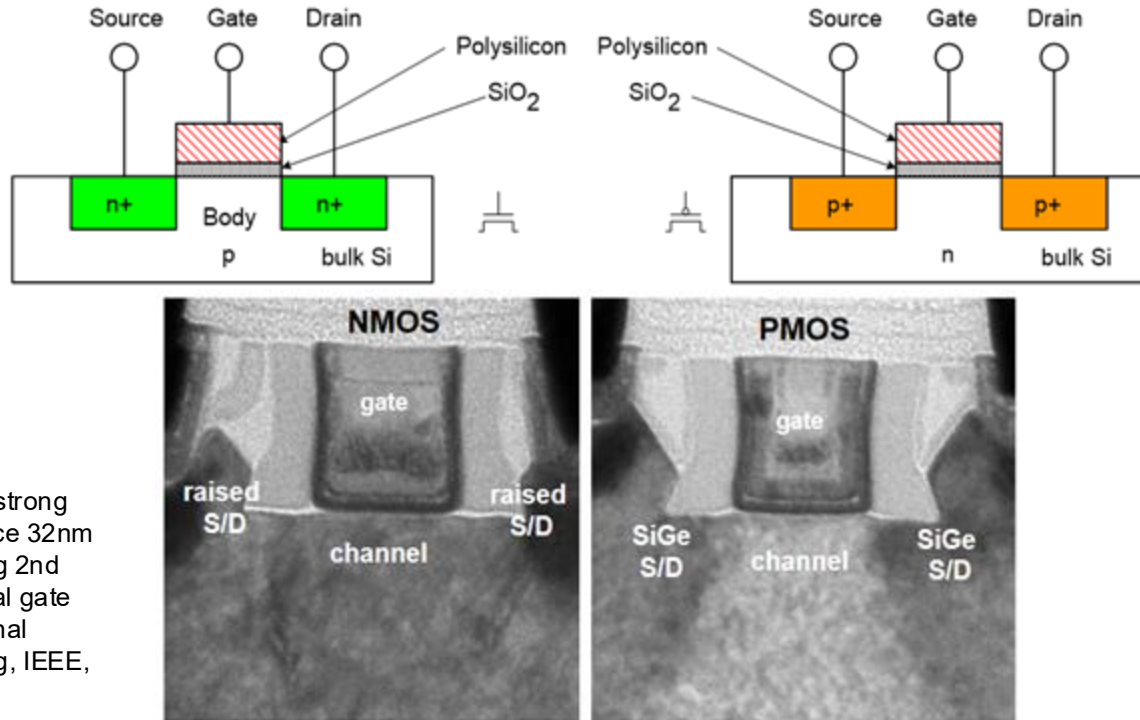
- Early commercial processes used only pMOS transistors and suffered from poor performance, yield, and reliability
- Processes using nMOS transistors became common in the 1970s
- While nMOS process was less expensive than CMOS, nMOS logic gates still consumed power while idle
- Power consumption became a major issue in the 1980s as hundreds of thousands of transistors were integrated onto a single die
- CMOS processes were widely adopted and replaced all the other technologies (pMOS, nMOS)

# Types of MOSFET

- pMOS, nMOS, CMOS
- Metal–insulator–semiconductor field-effect transistor (MISFET)
- Floating-gate MOSFET (FGMOS)
- Power MOSFET
- Multi-gate field-effect transistor (MuGFET)
- Quantum field-effect transistor (QFET)



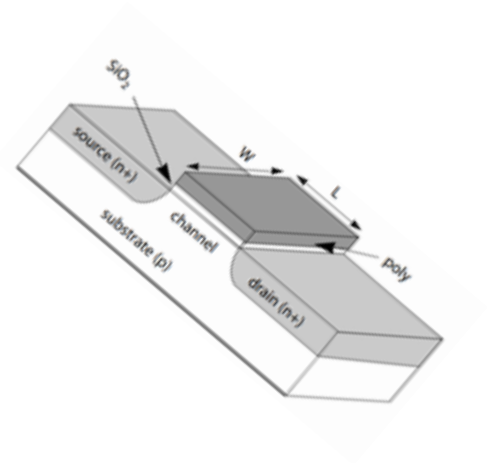
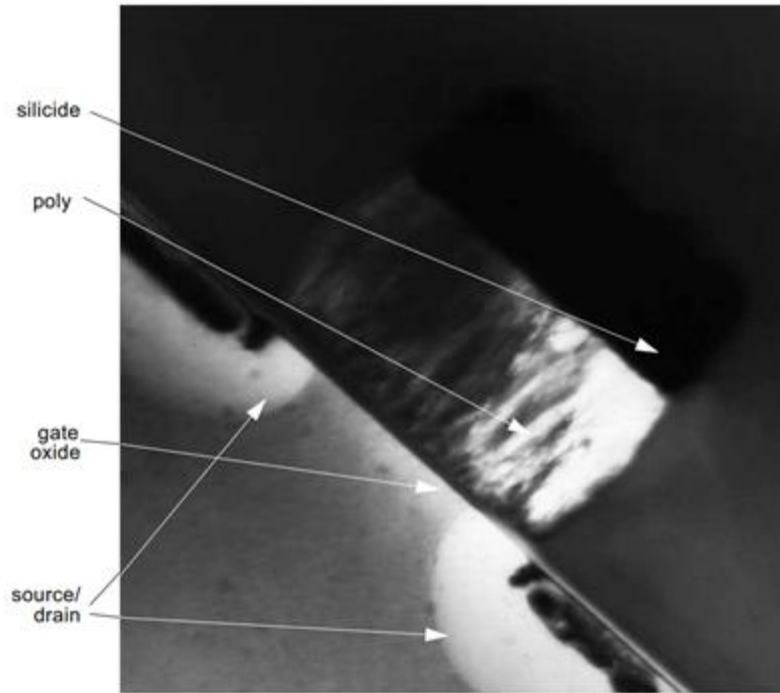
# A Closer Look



Packan P, Akbar S, Armstrong M, et al. High performance 32nm logic technology featuring 2nd generation High-k + metal gate transistors. In: International Electron Devices Meeting, IEEE, 2009. 659–662

Cross sectional view of intel's 32nm CMOS devices

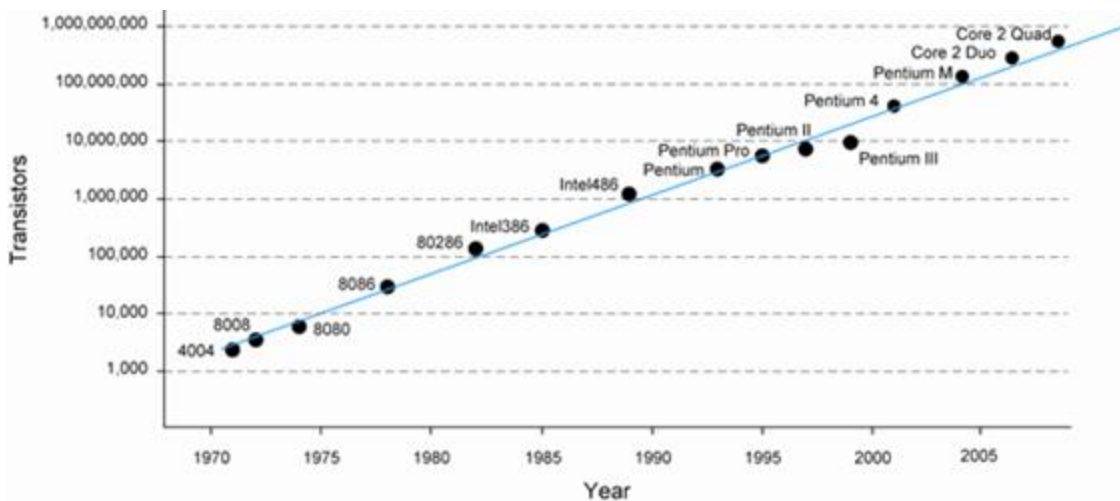
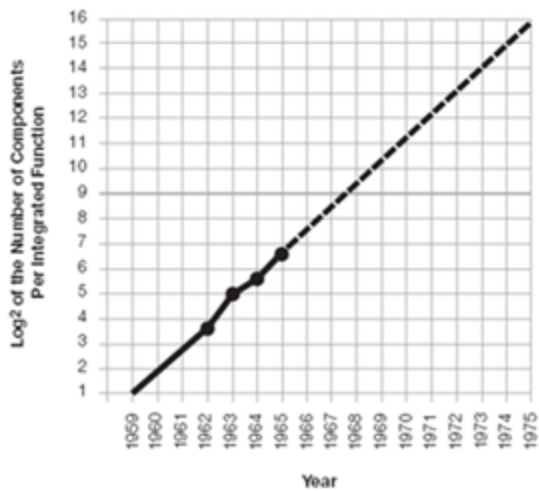
# A Closer Look



Photomicrograph of a submicron MOS transistor (courtesy Agere)

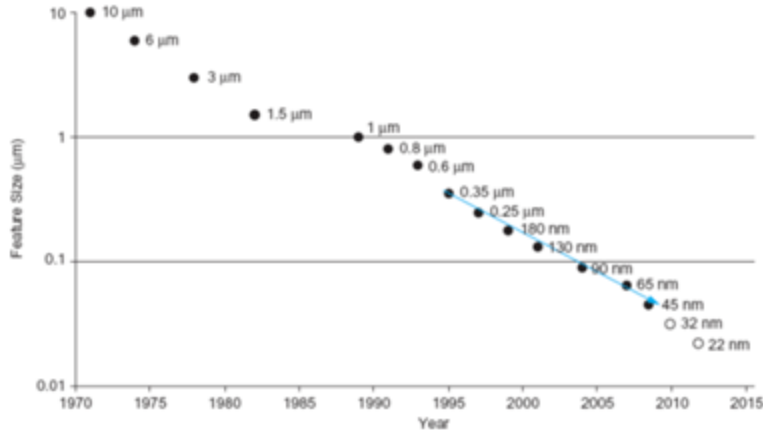
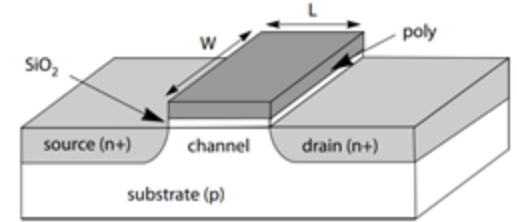
# Moore's Law

- 1965: Gordon Moore plotted # transistors on each chip vs year
  - Fit straight line on semi-log scale
  - Transistor counts have doubled about every 2 years



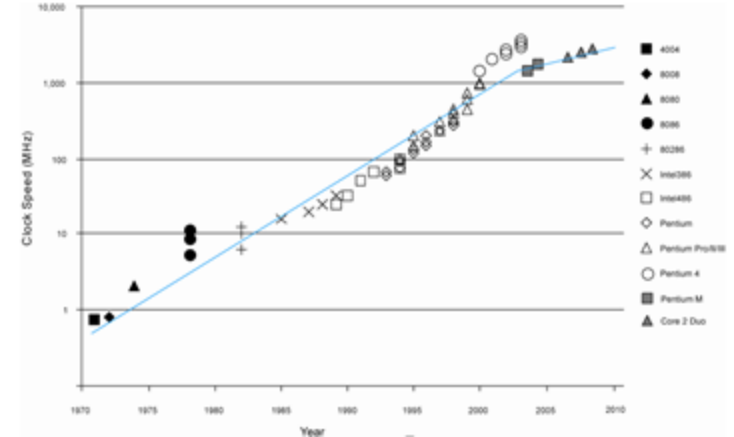
# Moore's Law

- Process generations
  - A manufacturing technology at a particular *channel length (feature size)* is called a **technology node/process node** (micron, submicron, nanometer technologies)



MOSFET scaling  
(process nodes)

|        |        |
|--------|--------|
| 10 μm  | – 1971 |
| 6 μm   | – 1974 |
| 3 μm   | – 1977 |
| 1.5 μm | – 1981 |
| 1 μm   | – 1984 |
| 800 nm | – 1987 |
| 600 nm | – 1990 |
| 350 nm | – 1993 |
| 250 nm | – 1996 |
| 180 nm | – 1999 |
| 130 nm | – 2001 |
| 90 nm  | – 2003 |
| 65 nm  | – 2005 |
| 45 nm  | – 2007 |
| 32 nm  | – 2009 |
| 22 nm  | – 2012 |
| 14 nm  | – 2014 |
| 10 nm  | – 2016 |
| 7 nm   | – 2018 |
| 5 nm   | – 2020 |



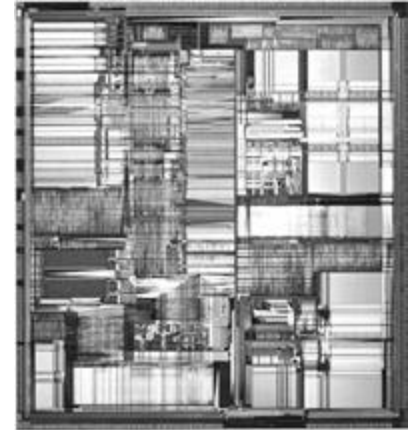
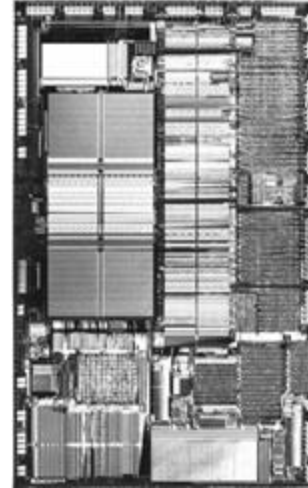
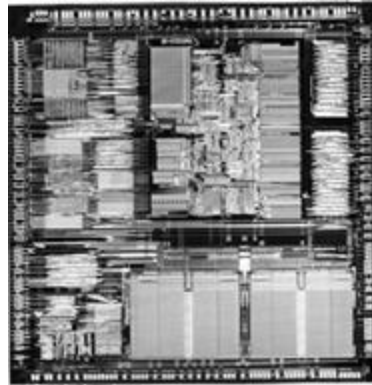
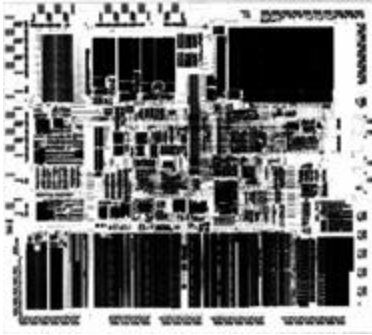
# Moore's Law

- Moore's law and intel microprocessors

| microprocessor                | date of introduction | # transistors |
|-------------------------------|----------------------|---------------|
| 80286                         | 2/82                 | 134,000       |
| 80386                         | 10/85                | 275,000       |
| 80486                         | 4/89                 | 1,200,000     |
| Intel Pentium™                | 3/93                 | 3,100,000     |
| Intel Pentium Pro™            | 11/95                | 5,500,000     |
| Intel Pentium II™             | 1997                 | 7,500,000     |
| Intel Pentium III™            | 1999                 | 9,500,000     |
| Intel Pentium 4™              | 2000                 | 42,000,000    |
| Intel Itanium™                | 2001                 | 25,000,000    |
| Intel Itanium 2™              | 2003                 | 220,000,000   |
| Intel Itanium 2™ (9 MB cache) | 2004                 | 592,000,000   |

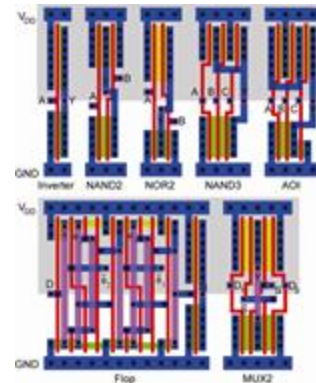
# Intel Processors over the years

- 80286, 80386, 80486, Pentium



# Types of chips

- Cell-based design
  - In the 1960s, standard parts were logic gates; in the 1970s they were LSI components. Today, standard parts include fairly specialized components: communication network interfaces, graphics accelerators
- Full custom
  - Designed manually piece by piece, fully custom layouts
- Application-specific integrated circuits (ASICs)
  - Rather than build a system out of standard parts, designers can now create a single chip for their particular application
- Systems-on-chips (SoCs)
  - a single-chip computer can include a CPU, bus, I/O devices, and memory

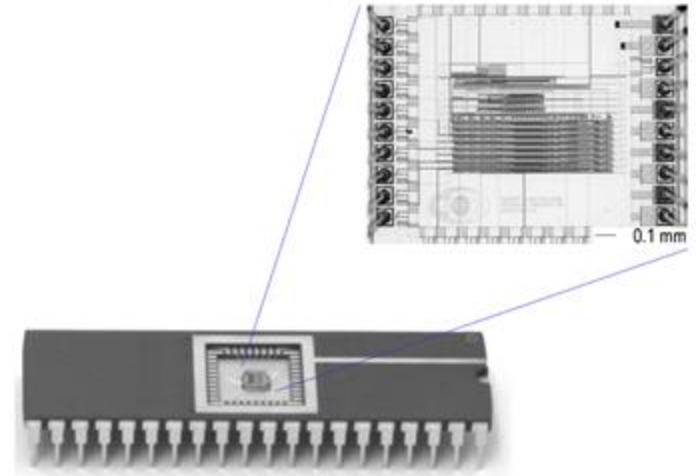
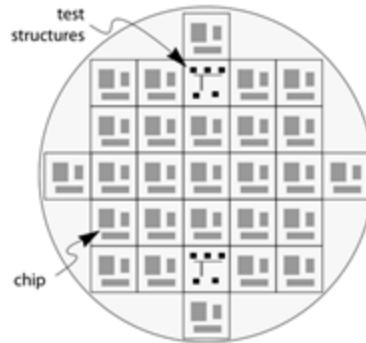
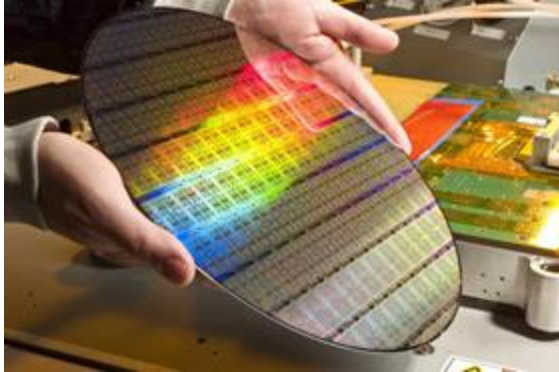


# VLSI Chip Types

- ❑ VLSI chips are classified by the approach used to implement and build the circuit
  - » **Full-custom Design**: where every circuit is custom designed for the project
    - Extremely tedious
    - Time-consuming process
  - » **Application-Specific Integrated Circuits (ASICs)**: using an extensive suite of CAD tools that portray the system design in terms of standard digital logic constructs
    - Including state diagrams, functions tables, and logic diagram
    - Designer does not need any knowledge of the underlying electronics or the physic of the silicon chip
    - Major drawback is that all characteristics are set by the architectural design
  - » **Semi-custom Design**: between that of a full-custom and ASICs
    - Using a group of primitive predefined cells as building blocks, called *cell library*

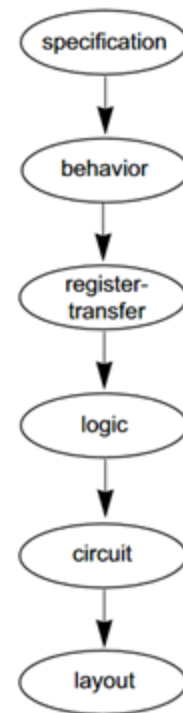
# Chip Lifecycle

- A wafer > fabrication > chips > packaging

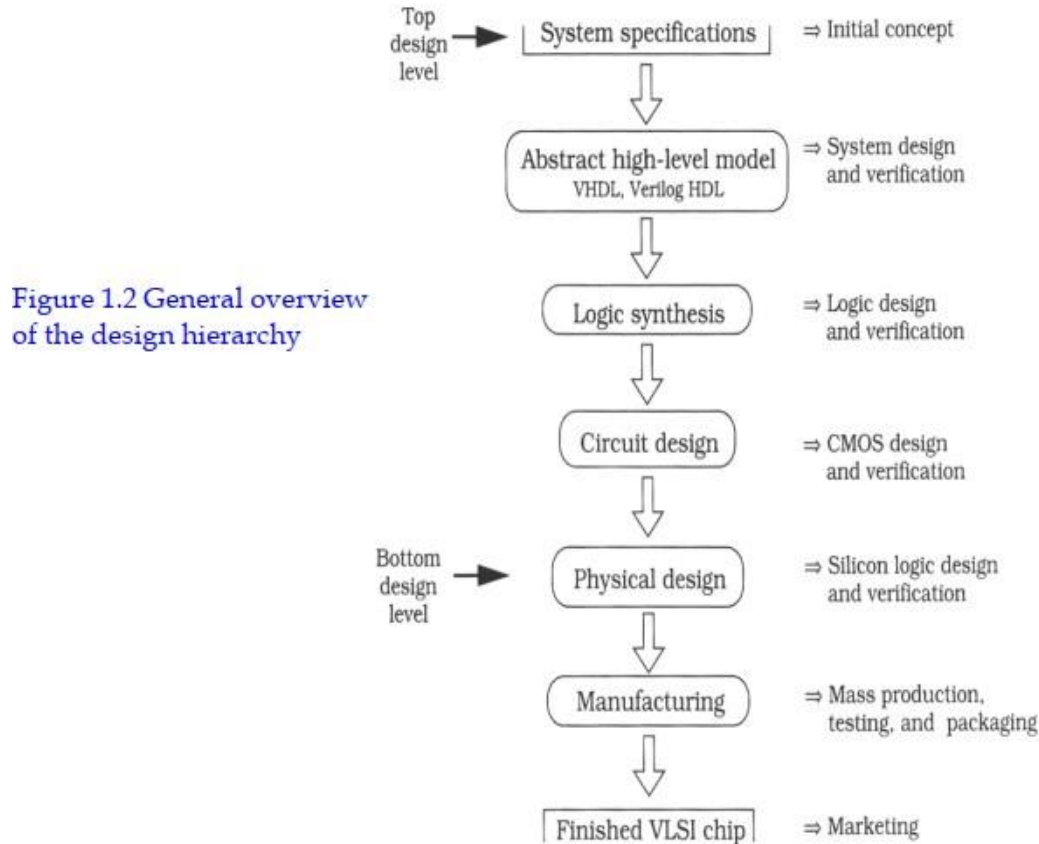


# Abstraction levels

- **Specification**
  - A set of requirements, not a design
- **Behavior**
  - generally modeled as some sort of executable program
- **Register-transfer**
  - The system is specified as Boolean functions stored in abstract memory elements
- **Logic**
  - The system is designed in terms of Boolean logic gates, latches, and flip-flops
- **Circuit**
  - The system is implemented as transistors
- **Layout**
  - The final design for fabrication



# Design Hierarchy (1/2)



# A Simple Design Flow

- An example of a design hierarchy
  - – an instruction design of a m

- $Register\_X \square A \square B$

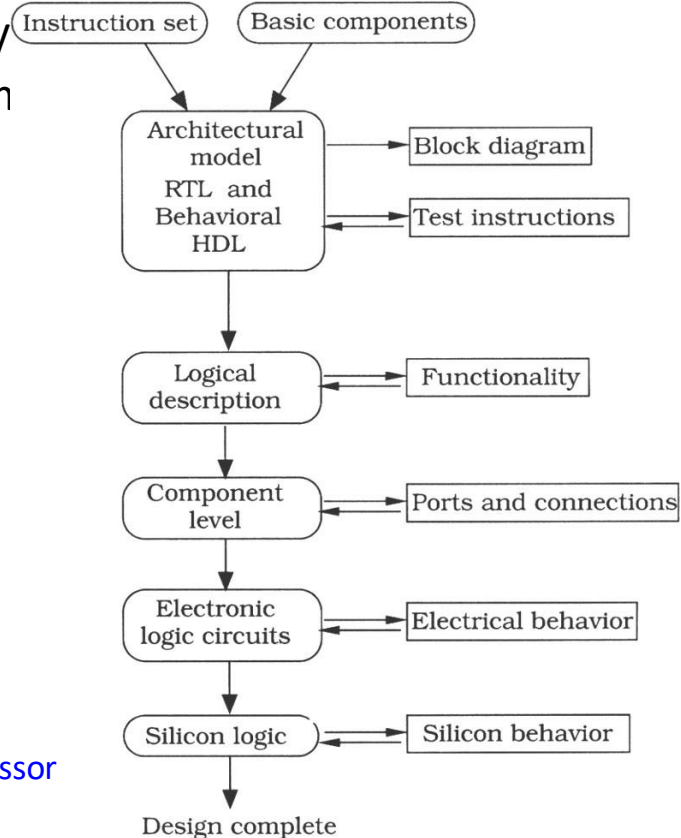


Figure 1.3 A simple design flow for a microprocessor

Reading from the text book.....

# Design Hierarchy (2/2)

## □ Hierarchical design

### » Top-down design

- the initial work is quite abstract and theoretical and there is no direct connection to silicon until many steps have been completed

### » Bottom-up design

- starts at the silicon or circuit level and builds primitive units such as logic gates, adders, and registers as the first steps

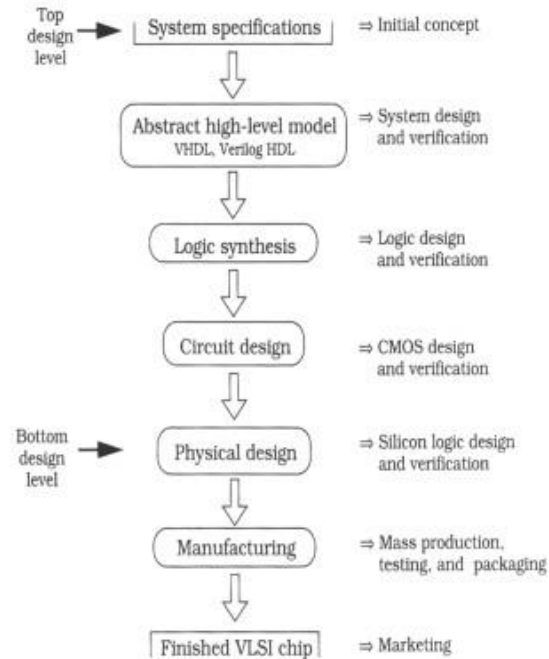
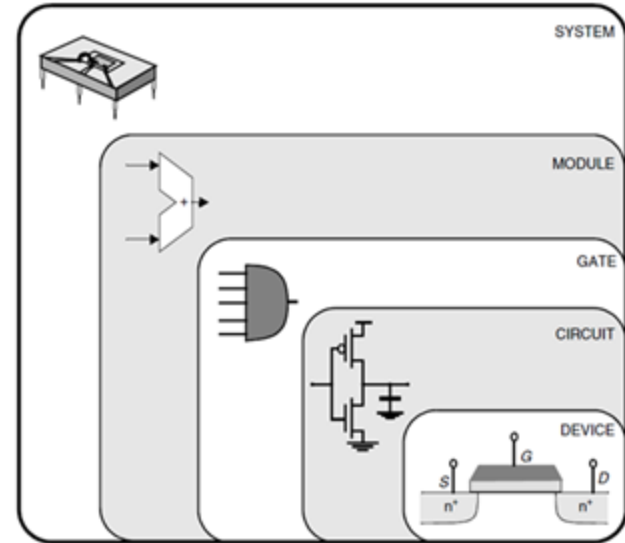


Figure 1.2 General overview of the design hierarchy

# Design methodologies

- Top-down approach
  - Start from the specification and reach device specification
- Bottom-up approach
  - Start from the device optimization and go up to meet specification



# Text

| # | Title                                             | Author(s)                             | Edition     |
|---|---------------------------------------------------|---------------------------------------|-------------|
| 1 | CMOS VLSI Design                                  | N.H.E. Weste, D. Harris & A. Banerjee | 4th ed.     |
| 2 | Fundamentals of Digital Logic with Verilog Design | Stephen Brown & Zvonko Vranesic       | 2nd/3rd ed. |

Transistors = Switches

# Slide References

1. <https://en.wikipedia.org/wiki/MOSFET>
2. <https://www.techspot.com/article/1840-how-cpus-are-designed-and-built-part-3/>
3. <https://ieeexplore.ieee.org/document/5424253>
4. Wayne Wolf - Modern VLSI Design IP based design
5. Jan M. Rabaey - Digital Integrated Circuits
6. Weste and Harris - CMOS VLSI Design

Thank you!